

Abstract

This paper proposed an approach that integrates both **software** and **hardware** techniques to uncover key properties of SMR drivers. The software component analyzes the latency of specific I/O operations, such as type, structure, and size of the persistent cache; type of cleaning algorithm; type of block mapping; and size of bands. The hardware component uses a high-speed camera to monitor the movement of the driver's read/write head through an observation window.

This paper also demonstrates the versatility and effectiveness of this method by applying it to three simulated and two real SMR drives, uncovering valuable details about their performance characteristics.

Background

Shingled Magnetic Recording (SMR) technology was introduced to overcome the superparamagnetic limit of Perpendicular Magnetic Recording (PMR) in achieving higher areal density. SMR increases storage density by narrowing track width and overlapping tracks, like shingles on a roof. However, this approach introduces significant data management challenges, as rewriting a single sector affects overlapping tracks, necessitating extensive Read-Modify-Write (RMW) operations.

To address these challenges, SMR employs the Shingle Translation Layer (STL), which abstracts the underlying complexities and ensures compatibility with existing file systems. STL typically manages the mapping between Logical Block Addresses (LBAs) and Physical Block Addresses (PBAs) using static or dynamic mapping and utilizes a persistent cache to store recently written data, reducing the performance impact of frequent writes. Additionally, STL employs lazy or aggressive cleaning strategies to optimize the cleaning process: lazy cleaning runs when cache space is low, while aggressive cleaning occurs during idle times. Despite these mechanisms improving storage efficiency, performance bottlenecks remain evident in high random write workloads, particularly due to frequent RMW operations.

Test Drives

This section mentions the use of both emulated and real drives in the study. Two types of STLs, **set-associative** and **fully associative**, were implemented on CMR drives to mimic SMR behavior, with configurations varying in persistent cache and cleaning strategies. These setups were validated through rigorous testing, including system booting with a Linux kernel. Additionally, tests were conducted on two real SMR drives and a comparable CMR drive, demonstrating consistent performance characteristics across different capacities.

Characterization tests

Since this section is quite long, I focus on a few points that I find particularly noteworthy to discuss:

Characterization Goals

The authors outline the primary goals of drive characterization, including identifying the drive type (SMR or CMR) and the type of persistent cache (disk or flash). These characteristics directly impact performance in areas such as random writes, sequential reads, and cleaning efficiency. The authors designed tests to analyze cleaning, cache configurations, and mapping structures.

Test Mechanisms

The Skylight test methodology combines fio microbenchmarks with hardware-level head tracking, using a camera to record head positions for precise performance data. This approach effectively resolves ambiguities in latency measurements and uncovers drive behaviors that cannot be observed through software testing alone.

Drive Type and Persistent Cache Type

The authors differentiate SMR and CMR drives by analyzing random write behaviors and examine the impact of persistent cache configurations. By testing the fixed nature of random write latencies, they confirm that SMR drives write sequentially to the cache and that latency varies depending on the cache structure and LBA range.

Disk Cache Location and Layout

The authors investigate the location and layout of the disk cache by leveraging the fragmented reads phenomenon. By analyzing seek time variations across different LBA ranges, the authors identify whether the cache is located at the Outer Diameter (OD), Inner Diameter (ID), or Middle Diameter (MD). The results provide insights into the impact of cache placement on read performance.

Cleaning Algorithm

The authors design a multi-step test that analyzes cleaning algorithm after pauses. The results reveal that Seagate-SMR drives perform aggressive cleaning during pauses. The authors also propose a hypothesized cleaning algorithm and validate it through further tests, highlighting its efficiency in maintaining performance.

Persistent Cache Size

The authors measure the persistent cache size by filling it with random writes and triggering the cleaning process. It distinguishes between raw cache size and mapped cache size and verifies the behaviors of SMR drives through experiments.

Band Size

The band size is estimated using the granularity of the cleaning process. Tests determine the band size at various offsets in SMR drives and analyze head movement during the cleaning process.

Effect of Mapping Type on Drive Reliability

Static mapping enhances writing performance but introduces risks of data loss during power failures. Simulations and tests highlight the importance of mapping type and cache configurations in ensuring reliability.

Zone Structure

Sequential read tests reveal the zone structure of SMR drives, showing horizontal zones composed of thousands of tracks. Unlike the vertical zones of CMR drives, horizontal zones may serve as the foundation for allocating isocapacity bands and static mapping.

Conclusions

The Skylight methodology effectively characterizes SMR drives, revealing that disabling volatile cache leads to higher write latency and reduced sequential throughput, while enabling it or using high queue depths significantly boosts random I/O performance. Cleaning processes require 0.6–45 seconds and start after ~1 second of idle time. These drives are well-suited for applications with high spatial locality and sufficient idle time for background cleaning. Future work should explore design improvements for better performance.

My thought

This paper effectively utilizes the Skylight methodology to uncover critical insights into SMR drive behavior, combining hardware and software techniques in a novel and detailed manner. The approach to reverse-engineering and characterize SMR drives without vendor information is commendable, showcasing practical tests and robust analysis.